

AKITA, Japan

WMO: 475820

Lat: 39.7172N

Lon: 140.0994E

Elev: 72

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	25.0	13.0	10.7	29.4	15.0	11.8	30.2	32.1	31.6	29.5	31.2	11.8	310	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.3	(c) 89.3	(d) 76.1	(e) 86.7	(f) 75.0	(g) 84.1	(h) 74.0	(i) 78.0	(j) 85.1	(k) 76.9	(l) 84.1	(m) 75.7	(n) 82.3	(o) 10.2	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.0	136.4	81.7	74.7	130.4	80.9	73.6	125.4	80.2	41.5	85.1	40.5	84.3	39.3	82.5	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.2	(b) 23.5	(c) 20.6	(e) 19.5	(f) 94.4	(g) 2.2	(h) 2.9	(i) 17.9	(j) 96.5	(k) 16.6	(l) 98.2	(m) 15.4	(n) 99.9	(o) 13.8	(p) 102.0		
			DB	WB	18.2	79.3	2.1	1.4	16.7	80.3	15.4	81.2	14.2	81.9	12.7	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	32.6	33.4	39.2	49.1	59.7	67.4	74.8	77.7	70.1	58.5	46.9	37.2
	DBStd	16.47	4.06	4.66	5.24	6.04	5.56	4.50	4.42	4.35	5.62	5.30	6.11	5.01
	HDD50	1964	541	464	338	88	2	0	0	0	0	4	129	398
	HDD65	5003	1006	883	801	477	181	26	0	0	15	209	543	862
	CDD50	3424	0	0	2	61	302	523	769	858	603	268	37	1
	CDD65	989	0	0	0	0	16	99	304	393	168	9	0	0
	CDH74	6684	0	0	0	4	99	389	1953	3315	909	15	0	0
	CDH80	1784	0	0	0	0	11	60	506	1004	202	1	0	0
Wind	WSAvg	9.6	11.5	11.0	11.0	10.3	9.2	8.1	7.7	8.0	8.0	8.6	10.0	11.3
Precipitation	PrecAvg	68.9	5.0	3.8	3.9	4.9	4.8	4.7	7.5	7.3	6.8	6.3	7.2	6.5
	PrecMax	93.5	8.4	8.2	7.3	15.7	11.4	14.0	21.4	17.8	14.5	11.3	14.4	12.2
	PrecMin	48.6	2.0	2.0	1.9	0.8	1.4	0.7	0.6	0.6	2.1	1.4	2.6	3.1
	PrecStd	9.9	1.5	1.3	1.2	2.1	1.9	2.6	3.9	3.7	2.8	2.1	2.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	51.3	60.2	72.8	81.2	85.8	92.8	93.9	90.2	75.7	66.2	56.0
		MCWB	42.8	44.4	47.9	56.0	62.1	69.4	76.6	77.1	75.7	65.2	58.2	51.7
	2%	DB	43.3	46.6	54.7	67.3	76.6	81.2	88.2	90.2	85.0	72.5	62.5	51.2
		MCWB	38.9	41.2	46.7	54.0	61.1	67.0	75.5	76.6	74.0	63.1	55.8	46.8
	5%	DB	40.6	43.5	51.3	63.3	73.2	78.4	85.1	87.7	81.7	70.2	59.6	48.2
		MCWB	36.5	38.4	45.4	51.7	59.7	66.4	74.1	75.9	72.0	61.7	53.8	44.0
	10%	DB	38.6	40.9	48.4	59.4	69.7	75.6	82.4	85.4	78.9	67.9	57.0	45.2
		MCWB	34.6	36.3	43.3	50.7	58.8	65.3	73.2	75.4	70.4	60.0	51.8	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	46.3	50.8	59.1	65.6	72.1	78.5	79.8	77.4	68.9	60.5	52.5
		MCDB	46.1	49.4	56.0	67.2	74.1	77.5	86.3	87.2	85.4	73.1	64.0	54.9
	2%	WB	39.8	42.2	48.6	56.0	63.6	70.1	77.1	78.5	75.3	65.7	57.5	47.4
		MCDB	42.5	45.6	53.0	63.4	71.2	75.7	84.7	85.4	82.5	70.1	60.9	50.5
	5%	WB	37.3	39.1	46.2	53.9	61.9	68.9	76.0	77.6	73.7	63.4	54.9	44.2
		MCDB	39.7	42.7	50.5	60.1	69.3	74.5	82.7	84.6	80.4	68.3	58.7	47.6
	10%	WB	35.2	36.5	43.7	52.1	60.3	67.4	74.8	76.7	71.9	61.2	52.3	41.3
		MCDB	38.0	40.2	48.2	58.1	67.5	73.3	80.8	83.6	78.0	66.3	56.2	44.7
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.8	12.2	14.5	14.1	12.6	11.0	12.3	13.5	14.4	12.6	9.4
		MCDBR	11.4	14.4	17.7	21.5	21.1	17.8	15.9	15.5	15.5	16.7	16.8	14.4
		MCWBR	9.2	11.1	12.5	11.5	9.9	8.0	6.9	6.6	7.3	9.9	12.5	12.2
	5% WB	MCDBR	11.6	13.7	15.8	16.0	15.7	12.8	12.2	12.4	13.0	13.2	14.8	13.9
		MCWBR	10.3	12.0	12.9	10.6	9.1	6.9	5.8	6.1	7.2	9.9	13.0	12.7
Clear-Sky Solar Irradiance	taub	0.399	0.438	0.508	0.549	0.561	0.537	0.546	0.513	0.455	0.430	0.422	0.408	
	taud	1.980	1.896	1.812	1.789	1.829	1.970	2.012	2.115	2.233	2.172	2.091	2.018	
	Ebn at Noon	226	236	233	235	235	241	238	242	249	239	219	212	
	Edh at Noon	41	51	62	67	66	57	55	48	40	38	36	36	
All-Sky Solar Radiation	RadAvg	408	658	1023	1412	1583	1612	1417	1465	1207	902	541	363	
	RadStd	32	79	92	142	166	145	171	166	90	67	54	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	+124
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page